

A Masked-Perturbation Model for Destabilization Security Games in Feedback Systems

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Motivation

Why this problem matters

- Feedback is essential for stabilizing engineered systems, but it also creates intrinsic destabilization vulnerabilities (IDVs).
- Destabilization attacks aim to drive the closed loop out of its stable operating regime.
- The practical question here is how to defend a fixed plant when security resources are limited.

Research Gap and Goal

The practical defense question

- Earlier versions of the Feedback Mitigation Problem focused on changing realizations.
- Those solutions can be hard to interpret and hard to implement in hardware.
- Goal: build an interpretable, budget-aware defense-allocation model grounded in robust-control vulnerability analysis.

Contributions

What this work adds

- A masked-perturbation model coupling attacker perturbations with defender masking actions.
- Pure- and mixed-strategy games centered on destabilization.
- A signal-level, budgeted defense model with operational meaning.
- A worked example showing dominance reduction and a mixed Nash equilibrium.

Background

From Vulnerability Analysis to Defense Design

Intrinsic destabilization vulnerability (IDV) analysis is built in three stages:

1. System modeling

- Isolate Δ
- Derive the model map M

2. Threat modeling

- Specify read/write access
- Specify budget and coordination
- Identify admissible Δ s

3. Vulnerability analysis

- Find the smallest destabilizing admissible perturbation
- Summarize exposed configurations

Structured Vulnerability Metric

The vulnerability is tied to the smallest admissible destabilizing perturbation.

$$\mu_{\Delta}(M) = \frac{1}{\inf_{\Delta \in \underline{\Delta}} \{\sigma_{\max}(\Delta) : \det(I - M\Delta) = 0\}}$$

- Smaller destabilizing perturbations imply greater intrinsic vulnerability.
- With structure, the structured singular value is the appropriate tool.

Prior Formulation: Feedback Mitigation Problem

- Classical FMP asks for an optimal realization that minimizes vulnerability under a threat model.
- That is analytically powerful, but not ideal when the administrator cannot change the architecture.
- This paper instead reallocates limited defenses over an exposed attack surface.

Masked-Perturbation Model

Core MPM Formulation

$$r = Mw + d_1, \quad w = [\Delta \circ \nabla]r + d_2$$

$$U_M(\Delta, \nabla) = \begin{cases} u_a(\Delta, \nabla), & \|F_l(M, \Delta \circ \nabla)\|_{H_\infty} = \infty \\ u_d(\Delta, \nabla), & \|F_l(M, \Delta \circ \nabla)\|_{H_\infty} < \infty \end{cases}$$

- Δ is the attacker perturbation; ∇ is the binary defense mask.
- With $u_a = 1$ and $u_d = 0$, the utility is just destabilization probability.

Why the Rank-1 Mask Matters

$$\nabla = \vec{\eta}_w \vec{\eta}_r^\top, \quad \Delta \circ \nabla = \text{Diag}(\vec{\eta}_w) \Delta \text{Diag}(\vec{\eta}_r)$$

- This makes a defense action correspond to securing signals, not arbitrary links.
- That makes the action space interpretable and physically meaningful.

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- **This also allows us to move the mask onto the system model:**

$$M \rightarrow \text{Diag}(\vec{\eta}_r) M \text{Diag}(\vec{\eta}_w)$$

Admissible Actions and Cost

$$C_{\nabla_i} = (\mathbf{1} - \vec{\eta}_r) \cdot \vec{c}_r + (\mathbf{1} - \vec{\eta}_w) \cdot \vec{c}_w$$

$$\inf_{\vec{d}} \sup_{\vec{a}} U(\vec{a}, \vec{d})$$

s.t. $C \leq C^*$

- The admissible sets come from the threat model and realizable security actions.
- Mixed strategies let the defender randomize while respecting a budget constraint.

Success Sets and Dominance

- A success set records which opponent actions are defeated by a given action.
- Weak domination means one action succeeds wherever another action succeeds.
- This reduces the strategic core before solving for equilibrium.

Mixed Strategies

$$U(\vec{a}, \vec{d}) = \sum_{i,j} a_i d_j U_M(\Delta_i, \nabla_j)$$

- The defender's mix can be read as a randomized patrol or monitoring schedule.
- That is a major advantage over abstract realization redesign.

Worked Example

Example System

$$M = \frac{1}{1 + 8PK} \begin{bmatrix} P & PK \\ -8PK & K \end{bmatrix}$$

- The example exposes a 2×2 read/write attack surface.
- The threat model then specifies three admissible perturbation operators.

Example Threat Model

Let our attack models be:

- $\Delta_1 = \begin{bmatrix} * & 0 \\ 0 & 0 \end{bmatrix}$
- $\Delta_2 = \begin{bmatrix} 0 & 0 \\ 0 & * \end{bmatrix}$
- $\Delta_3 = \begin{bmatrix} 0 & * \\ 0 & 0 \end{bmatrix}$

And our admissible defenses:

- $\nabla_1 = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$
- $\nabla_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$
- $\nabla_3 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$
- $\nabla_4 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

with the * entries designed according to [1] to be sufficient to destabilize the exposed system from that access point. For the full transfer-function formulas, see the paper.

Budget-Constrained Defense Actions

Cost to Secure Signals:

- r_1 : 3
- r_2 : 1
- w_1 : 2
- w_2 : 1

Implication

- The full attack surface cannot be secured.
- Only four nonempty defense masks are admissible.
- This forces a nontrivial allocation problem.

Total Defense Budget: $C^* = 2$

Success Sets and Dominance Reduction

Success sets are defined as:

If the player chooses this actions, against what counteractions do they win?

$$\begin{aligned}\mathcal{S}(\Delta_1) &= \{\nabla_1, \nabla_2, \nabla_4\}, & \mathcal{S}(\nabla_1) &= \{\Delta_2, \Delta_3\}, \\ \mathcal{S}(\Delta_2) &= \{\nabla_3\}, & \mathcal{S}(\nabla_2) &= \{\Delta_2\}, \\ \mathcal{S}(\Delta_3) &= \{\nabla_2\}, & \mathcal{S}(\nabla_3) &= \{\Delta_1, \Delta_3\}, \\ & & \mathcal{S}(\nabla_4) &= \{\Delta_2, \Delta_3\}.\end{aligned}$$

- In the example, Δ_1 dominates Δ_3 , as $\mathcal{S}(\Delta_3) \subseteq \mathcal{S}(\Delta_1)$.
- Likewise, ∇_4 dominates both ∇_1 and ∇_2 .
- The game therefore reduces from a 3×4 action space to a 2×2 game.

Reduced Game and Mixed Equilibrium

$$0.5\Delta_1 \oplus 0.5\Delta_2, \quad 0.5\nabla_3 \oplus 0.5\nabla_4$$

- With $u_a = 1$ and $u_d = 0$, the expected utility is 0.5.
- Each side wins about half the time in the reduced game.

This is strictly an improvement on the undefended system, where any of the attacks is successful 100% of the time, and over any static defense, which also has 100% failure if the attacker knows the deployed defense strategy.

Why the Mixed Strategy Is Interpretable

- Randomized guard-post changes.
- Inspection software rotating across substations.
- Monitoring resources shifted across exposed servers.

The model produces an executable policy, not just an abstract operator-level optimum.

Discussion and Outlook

Open Problems

Attacker side

- Simultaneous destabilization against multiple masks.
- Structural conditions based on shared footprints.
- Connection to simultaneous stabilization, but reversed.

Defender side

- Stackelberg security games.
- Defender commits first, attacker observes.
- New algorithmic and observability questions.

Takeaways

Security in feedback systems can be formulated as a game between destabilizing perturbations and masking defenses.

- MPM turns IDV analysis into a budgeted resource-allocation problem.
- Rank-1 masks preserve interpretability at the signal level.
- Dominance reduction and mixed strategies yield simple executable defense policies.
- Open problems remain on simultaneous destabilization and Stackelberg variants.

Future Work

Ph.D. Dissertation Wish List

How This Continues the Main Story

- My thesis fixed the plant and optimized *coverage* over exposed read/write signals.
- The next questions ask what happens when we strengthen either player: richer attacks, richer defenses, or more freedom in the realization itself.
- Conceptually, the thread stays the same: use structure to turn abstract security questions into concrete design choices.

Future Work: Stronger Attacker Models

- **Simultaneous destabilization:** can one admissible minimal attack destabilize several (non-subset) defense masks at once?
- **Structure questions:** when do subset or overlap relations between masks imply dominance of one attack over another?
- **Algorithmic direction:** build a reduced action set, then optimize attack mixing over that set.

Future Work: Stronger Defense Design

- Alternative objectives: minimize defense cost for a target expected utility, not only maximize security under a fixed budget.
- Search questions: can we find sub-exponential or approximate algorithms for useful masks or useful mixed defenses?
- Strategic question: can a defense be chosen to force the attacker to mix over more actions and thereby improve stability odds?

Reconsidering the Full FMP

$$\begin{bmatrix} y(s) \\ r(s) \end{bmatrix} = \begin{bmatrix} G_{yu} & G_{yw} \\ G_{ru} & G_{rw} \end{bmatrix} \begin{bmatrix} u(s) \\ w(s) \end{bmatrix}$$

- In the full problem, the system behavior G_{yu} is fixed, but the implementation may vary.
- A naive solution would “hide” vulnerable dynamics from exposed signals.
- A better formulation randomizes the attack surface itself and minimizes *expected* vulnerability over that distribution.

Partial Work: Original FMP Formulation

$$\min_{Q \in \mathcal{Q}} \max_{\Delta \in \Delta} \left\| \Gamma(I - Q)^{-1} \Delta_a \right\|$$

- This formulation optimizes over subsystem implementations rather than over masks.
- Partial results include vulnerability bounds and two approximation directions: a **Hilbert–Frobenius projection** approach and a **linear-program** search over parameterized realizations.
- The main challenge is that the optimization is generally non-convex and structurally constrained.

Nonlinear Extension

For a system $(\dot{x}, y, r) = f(x, u, w)$, $w = \Delta r$:

- Bound stacked input $[u, w]$ s.t. $w \in \mathcal{W}$
- Prove an ultimate uniform bound on r , i.e. $w \in \mathcal{W} \implies r \in \mathcal{R}$.
- Δ operators that satisfy the feedback remaining in the proven safe set, i.e., $r \in \mathcal{R} \implies \Delta r \in \mathcal{W}$.

Then the feedback connection $w = \Delta r$ is stable.

What I've Done to Prepare

- I have a draft script (WIP) to demonstrate the vulnerability bounds and the two full FMP design algorithms found in the appendix of the Thesis.
- I have a detailed implementation plan for each of my future work questions, found in the following GitHub repository:
- https://github.com/perryGabriel/masked-perturbation-model/blob/main/docs/masked_perturbation_future_work_plan.md.

Thank you

Questions?